

1.10

Specify all the relationships among the records of the database shown in Figure 1.2.

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

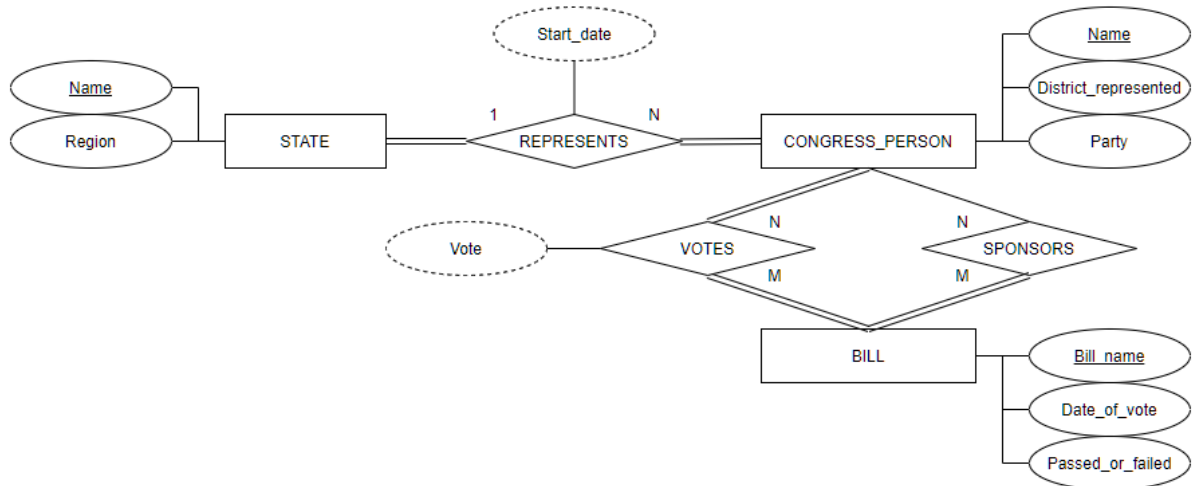
- GRADE_REPORT 中的 Student_number 為某個 STUDENT 的 Student_number
- GRADE_REPORT 的 Section_identifier 為某個 SECTION 的 Section_identifier
- PREREQUISITE 中的 Course_number 為某兩個 COURSE 的 Course_number
- SECTION 中的 Course_number 為某個 COURSE 的 Course_number

3.21

Design an ER schema for keeping track of information about votes taken in the U.S. House of Representatives during the current two-year congressional session. The database needs to keep track of each U.S. STATE's Name (e.g., 'Texas', 'New York', 'California') and include the Region of the state (whose domain is { 'Northeast', 'Midwest', 'Southeast', 'Southwest', 'West' }).

Each CONGRESS_PERSON in the House of Representatives is described by his or her Name, plus the District represented, the Start_date when the congressperson was first elected, and the political Party to which he or she belongs (whose domain is { 'Republican', 'Democrat', 'Independent', 'Other' }).

The database keeps track of each BILL (i.e., proposed law), including the Bill_name, the Date_of_vote on the bill, whether the bill Passed_or_failed (whose domain is { 'Yes' , 'No' }), and the Sponsor (the congressperson(s) who sponsored—that is, proposed—the bill). The database also keeps track of how each congressperson voted on each bill (domain of Vote attribute is { 'Yes' , 'No' , 'Abstain' , 'Absent' }). Draw an ER schema diagram for this application. State clearly any assumptions you make.



Entities :

- STATE：有名稱和區域的資料，Primary key 為 Name
- CONGRESS_PERSON：有姓名、代表區和政黨的資料，Primary key 為 Name
- BILL：有名稱、投票日和是否通過的資料，Primary key 為 Bill_name

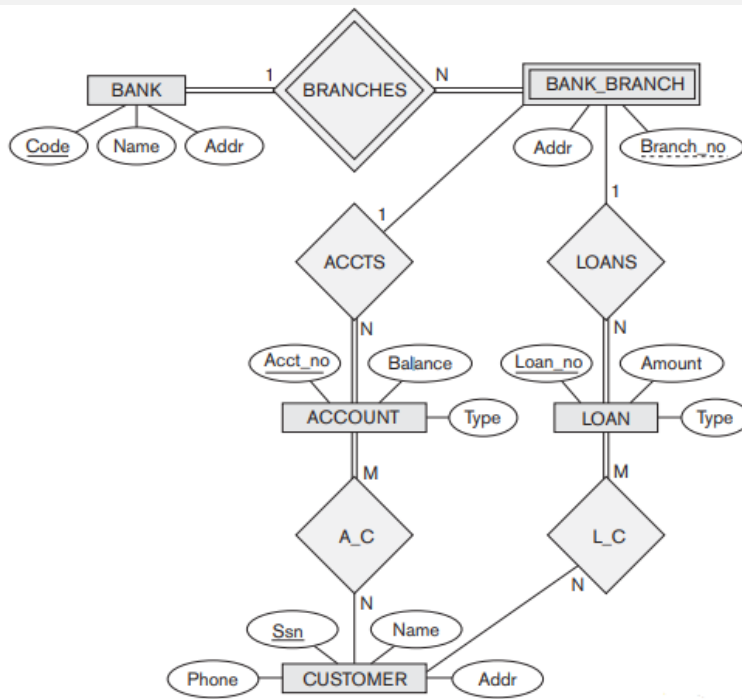
Relationships :

- REPRESENTS：每個 STATE 會有至少一個 CONGRESS_PERSON，每個 CONGRESS_PERSON 只代表一個 STATE；Start_date 為 CONGRESS_PERSON 開始工作的日期
- SPONSORS：每個 BILL 由至少一個 CONGRESS_PERSON 提出，但 CONGRESS_PERSON 不一定會提出 BILL
- VOTES：每個 CONGRESS_PERSON 不管有沒有投票都會記錄在 VOTES 中，而每個 BILL 都一定要被投票；Vote 為每個 CONGRESS_PERSON 的出席或投票情形

3.23

Consider the ER diagram shown in Figure 3.22 for part of a BANK database. Each bank can have multiple branches, and each branch can have multiple accounts and loans.

- List the strong (nonweak) entity types in the ER diagram.
- Is there a weak entity type? If so, give its name, partial key, and identifying relationship.
- What constraints do the partial key and the identifying relationship of the weak entity type specify in this diagram?
- List the names of all relationship types, and specify the (min, max) constraint on each participation of an entity type in a relationship type. Justify your choices.
- List concisely the user requirements that led to this ER schema design.
- Suppose that every customer must have at least one account but is restricted to at most two loans at a time, and that a bank branch cannot have more than 1,000 loans. How does this show up on the (min, max) constraints?



- BANK、ACCOUNT、LOAN、CUSTOMER
- Yes
 - Name: BANK_BRANCH
 - Partial key: Branch_no
 - Identifying relationship: BRANCHES
- 一對多且均為完全參與
- BANK----- $(1, N)$ --**BRANCHES**-- $(1, 1)$ -----BANK_BRANCH
 - ACCOUNT----- $(1, 1)$ --**ACCTS**-- $(0, N)$ -----BANK_BRANCH
 - LOAN----- $(1, 1)$ --**ACCTS**-- $(0, N)$ -----BANK_BRANCH
 - ACCOUNT----- $(1, N)$ --**A_C**-- $(0, N)$ -----CUSTOMER
 - LOAN----- $(1, N)$ --**L_C**-- $(0, N)$ -----CUSTOMER
- 每間銀行有名稱、地址和唯一的編號；每間銀行有0-N 個分行，分行有地址和分行編號。每個分行有多個存戶跟借貸戶；存戶和借貸戶都分別包含類型、金額和唯一的編號。存戶和借貸戶都被客戶所擁有，客戶則有姓名、地址、電話和唯一的 SSN 等資料。
- ACCTS**-- $(0, 1000)$ -----BANK_BRANCH
 - A_C**-- $(1, N)$ -----CUSTOMER
 - L_C**-- $(0, 2)$ -----CUSTOMER

4.20

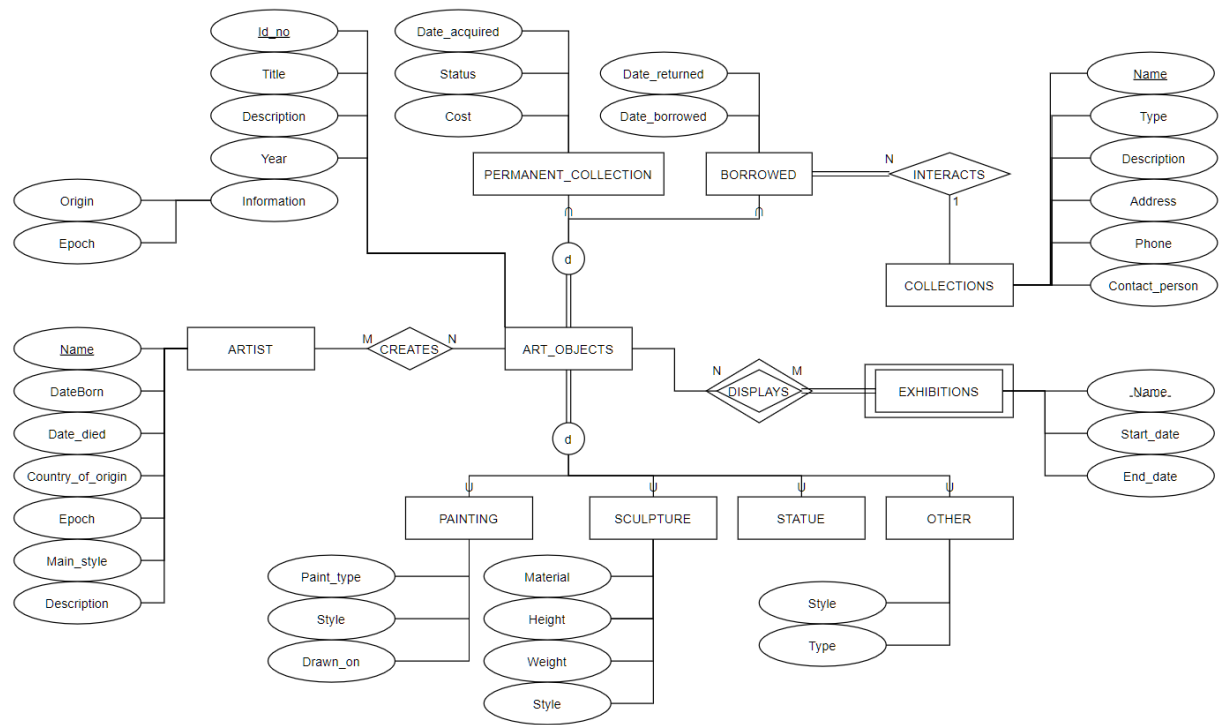
Design a database to keep track of information for an art museum. Assume that the following requirements were collected:

- The museum has a collection of ART_OBJECTS. Each ART-OBJECT has a unique Id_no, an Artist (if known), a Year (when it was created, if known), a

Title, and a Description. The art objects are categorized in several ways, as discussed below.

- ART_OBJECTS are categorized based on their type. There are three main types—PAINTING, SCULPTURE, and STATUE—plus another type called OTHER to accommodate objects that do not fall into none of the three main types.
- A PAINTING has a Paint_type (oil, watercolor, etc.), material on which it is Drawn_on (paper, canvas, wood, etc.), and Style (modern, abstract, etc.).
- A SCULPTURE or a statue has a Material form which it was created (wood, stone, etc.), Height, Weight, and Style.
- An art object in the OTHER category has a Type (print, photo, etc.) and Style.
- ART_OBJECTs are categorized as either PERMANENT_COLLECTION captured about objects in the PERMANENT_COLLECTION includes Date_acquired, Status (on display, on loan, or stored), and Cost. Information captured about BORROWED objects includes the Collection from which it was borrowed, Date_borrowed, and Date_returned.
- Information describing the country or culture of Origin (Italian, Egyptian, American, Indian, and so forth) and Epoch (Renaissance, Modern, Ancient, and so forth) is captured for each ART_OBJECT.
- The museum keeps track at ARTIST information, if known: Name, DateBorn (if known), Date_died (if not living), Country_of_origin, Epoch, Main_style, and Description. The Name is assumed to be unique.
- Different EXHIBITIONS occur, each having a Name, Start_date, and End_date. EXHIBITIONS are related to all the art objects that were on display during the exhibition.
- Information is kept on other COLLECTIONS with which the museum interacts; this information includes Name (unique), Type (museum, personal, etc.), Description, Address, Phone, and current Contact_person.

Draw an EER schema diagram for this application. Discuss any assumptions you make, and then justify your EER design choices.



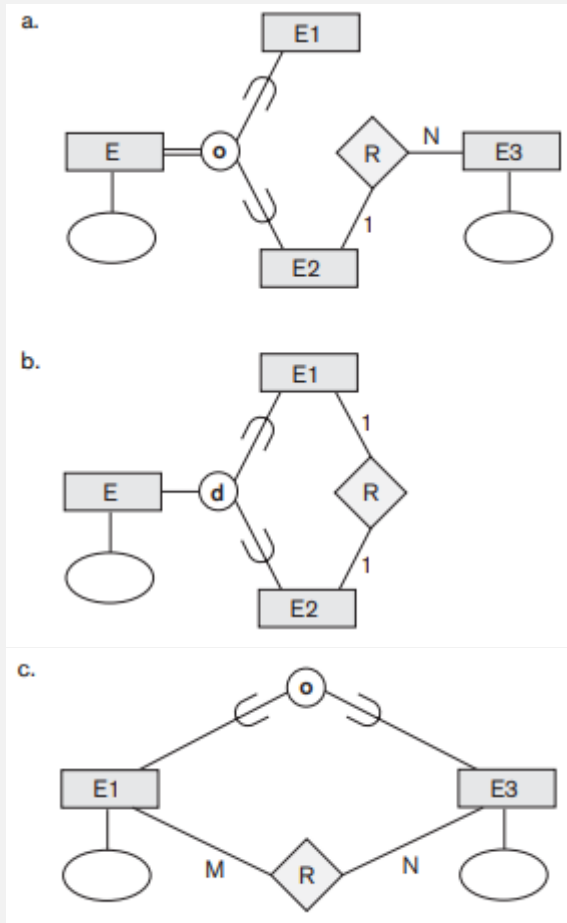
Entities :

- ART_OBJECTS
 - Primary key 為 Id_no，有 Title、Description 等資料，其中 Information 包含 Origin 和 Epoch。
 - PAINTING、SCULPTURE、STATUE、OTHER 繼承 ART_OBJECTS，代表不同類型的藝術品資料，並分別有自己的其它欄位
 - PERMANENT_COLLECTION 和 BORROWED 繼承 ART_OBJECTS，代表館藏和外借的藝術品資料，並分別有自己的其它欄位
- ARTIST: Primary key 為 Name，有生卒日等資料
- COLLECTIONS
 - Primary key 為 Name，有其它 collection 相關的資料。
- EXHIBITIONS
 - Partial key 為 Name，有展覽起始的時間。

Relationships :

- CREATES
 - 藝術家與藝術品的資料分別管理，有可能有藝術家的資料但是博物館沒有該藝術家的作品，也有可能藝術品但是藝術家不可考；一個藝術家可能有多件作品，一件作品可能有多位作者。
- INTERACTS
 - 博物館可能會將藝術品外借給其他館，一次只可能借給一個館
- DISPLAYS
 - 每件藝術品可能會在不同的展覽展出

4.26 Which of the following EER diagram is/are incorrect and why? State clearly any assumptions you make.



C 是錯誤的，應該還有一個 Entity 在 O 的上面，作為 E1和 E3繼承的對象。